Mohsen Zakeri

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Most Recent Position

Postdoctoral Fellow, Computer Science Department 2022 - 2025
Johns Hopkins University, Baltimore, MD

Advisor: Ben Langmead

Education

PhD | ***Computer Science, Advisor: Prof. Rob Patro*** Dec. 2021
University of Maryland College Park, MD

MS | ***Computer Science, Advisor: Prof. Rob Patro*** May. 2017
Stony Brook University Stony Brook, NY

BS | ***Computer Engineering, Major: Software Engineering*** Jun. 2015
University of Tehran Tehran, Iran

Research Experience

Scalable full-text pangenome indexes 2022 - present
Postdoc fellow at Johns Hopkins University

- Designed and developed Movi, a data structure based on the Move-Structure for indexing pangenomes. Movi is a full-text index which scales very well for highly similar references while being very fast to query. | *C++*

Oxford Nanopore reads classification 2022 - present
Postdoc fellow at Johns Hopkins University

- developed Movi Color for multi-class and taxonomic classification of metagenomic samples.
- Developed a method to compute exact matching queries (pseudo-matching lengths) with Movi for real-time classification of Oxford Nanopore reads and fast host depletion with pangenomes. | *C++*
- Contributed to the design of Sigmoni for multi-class classification of Nanopore signals using a compressed full-text index (r-index).

Efficient and accurate methods for single-cell RNA-seq preprocessing 2020 - 2021
Research assistant at University of Maryland

- Contributed to the design and development of a variation of pseudoalignment with structural constraints (sketch mode) for mapping single-cell RNA-seq reads, used in Alevin-fry. | *C++ and Rust*

A fast and accurate aligner for short reads 2019 - 2020
Research assistant at Stony Brook University and University of Maryland

- Collaborated on the design and development of Puffaligner, an aligner based on Pufferfish (a ccdbg index) that aligns various types of short reads to a collection of references. | *C++*

A pipeline for abundance estimation of metagenomics sequencing reads 2018 - 2019
Research assistant at Stony Brook University

- Collaborated on the design and development of Cedar, a fast and accurate method for quantifying metagenomics sequencing reads. Cedar focuses on eliminating spurious reads in an iterative manner during the EM optimization. | *C++*

Improving the accuracy of fast RNA-seq quantification methods 2016 - 2019
Research assistant at Stony Brook University

- Enhanced the probabilistic model of bulk RNA-seq in Salmon to improve the fidelity of the equivalence class-based factorization models for RNA-seq data. This new model improved the accuracy of abundance estimations, particularly in challenging cases, i.e., paralogous genes. | *C++*
- Collaborated on the design and development of Selective-alignment to enhance the specificity and sensitivity of lightweight alignment methods used in fast RNA-seq quantification approaches. | *C++*
- Improved the uncertainty estimation of missing data in RNA-seq quantification by using augmented-data bootstrap. This approach is specifically useful in the presence of allelic imbalance.

Peer Reviewed Publications

*: co-first authorship

- Majidian, S., Hwang, S., **Mohsen Zakeri**, & Langmead, B. (2025). Evani benchmarking workflow for evolutionary distance estimation. *Briefings in Bioinformatics*, *26*(3), bbaf267
- Guccione, C., Patel, L., Tomofuji, Y., McDonald, D., Gonzalez, A., Sepich-Poore, G. D., Sonehara, K., **Mohsen Zakeri**, Chen, Y., Dilmore, A. H., et al. (2025). Incomplete human reference genomes can drive false sex biases and expose patient-identifying information in metagenomic data. *Nature Communications*, *16*(1), 825
- **Mohsen Zakeri**, Brown, N. K., Ahmed, O. Y., Gagie, T., & Langmead, B. (2024). Movi: A fast and cache-efficient full-text pangenome index. *iScience*, *27*(12)
- Shivakumar, V. S., Ahmed, O. Y., Kovaka, S., **Mohsen Zakeri**, & Langmead, B. (2024). Sigmoni: Classification of nanopore signal with a compressed pangenome index. *Bioinformatics (ISMB Proceedings)*, *40*(Supplement_1), i287–i296
- Wu, E. Y., Singh, N. P., Choi, K., **Mohsen Zakeri**, Vincent, M., Churchill, G. A., Ackert-Bicknell, C. L., Patro, R., & Love, M. I. (2023). Seesaw: Detecting isoform-level allelic imbalance accounting for inferential uncertainty. *Genome biology*, *24*(1), 165
- He, D., **Mohsen Zakeri**, Sarkar, H., Soneson, C., Srivastava, A., & Patro, R. (2022). Alevin-fry unlocks rapid, accurate and memory-frugal quantification of single-cell rna-seq data. *Nature Methods*, *19*(3), 316–322
- Skoufos, G., Almodaresi, F., **Mohsen Zakeri**, Paulson, J. N., Patro, R., Hatzigeorgiou, A. G., & Vlachos, I. S. (2022). Agamemnon: An accurate metagenomics and metatranscriptomics quantification analysis suite. *Genome biology*, *23*(1), 1–27
- Almodaresi, F., **Mohsen Zakeri***, & Patro, R. (2021). Puffaligner: A fast, efficient, and accurate aligner based on the pufferfish index. *Bioinformatics*
- Srivastava, A., Malik, L., Sarkar, H., **Mohsen Zakeri**, Almodaresi, F., Soneson, C., Love, M. I., Kingsford, C., & Patro, R. (2020). Alignment and mapping methodology influence transcript abundance estimation. *Genome biology*, *21*(1), 1–29
- Sarkar, H., **Mohsen Zakeri***, Malik, L., & Patro, R. (2018). Towards selective-alignment: Bridging the accuracy gap between alignment-based and alignment-free transcript quantification, 27–36
- **Mohsen Zakeri**, Srivastava, A., Almodaresi, F., & Patro, R. (2017a). Improved data-driven likelihood factorizations for transcript abundance estimation. *Bioinformatics (ISMB Proceedings)*, *33*(14), i142–i151

Pre-prints

- **Mohsen Zakeri**, Brown, N. K., Gagie, T., & Langmead, B. (2025). Movi 2: Fast and space-efficient queries on pangenomes. *bioRxiv*, 2025–10
- Brown, N. K., Depuydt, L., **Mohsen Zakeri**, Alhadi, A., Allam, N., Begleiter, D., Karpagavalli, N. B. K., Khajjayam, S. S., Wahed, H., Gagie, T., & Langmead, B. (2025). Kebab: k-mer based breaking for finding super-maximal exact matches. *arXiv preprint arXiv:2502.20338*
- Tan, S., Majidian, S., Langmead, B., & **Mohsen Zakeri**. (2025). Movi color: Fast and accurate long-read classification with the move structure. *bioRxiv*, 2025–05
- Srivastava, A., **Mohsen Zakeri**, Sarkar, H., Soneson, C., Kingsford, C., & Patro, R. (2021). Accounting for fragments of unexpected origin improves transcript quantification in rna-seq simulations focused on increased realism. *bioRxiv*
- **Mohsen Zakeri**, Srivastava, A., Sarkar, H., & Patro, R. (2021). A like-for-like comparison of lightweight-mapping pipelines for single-cell rna-seq data pre-processing. *bioRxiv*

Conferences and Presentations

- **Sequitur: A space-efficient and versatile method for k-mer queries on pangenomes (poster)**. In: In *Biological data science, cold spring harbor, ny.* 2024
- **Movi: a fast and cache-efficient full-text pangenome index (talk)**. In: In *Recomb-seq, cambridge, ma.* 2024
- **Real-time nanopore adaptive sampling with Movi (talk)**. In: In *Genome informatics, cold spring harbor, ny.* 2023
- **Real-time nanopore adaptive sampling with move data structure (talk)**. In: In *Workshop on emerging methods for sequencing analysis, state college, pa.* 2023
- **Augmented-data bootstrap reveals missing uncertainty during RNA-seq quantification (poster)**. In: In *Recomb, san diego.* 2022
- **Cedar: scalable, accurate and fast metagenomic abundance estimation (poster)**. In: In *Recomb, washington d.c.* 2019

Professional Experience

Postdoctoral Fellow 2022 - 2025
Johns Hopkins University - Langmead lab Baltimore, MD

- Involved in projects focused on design and developtment of efficient data structures for pangenomes and Oxford Nanopore read classification. | *C++*

Computational Biologist Intern Summer 2021
Ocean Genomics Pittsburgh, PA

- Developed Mudskipper, a software tool that converts alignments in SAM files - commonly used for storing sequence alignments - from genomic coordinates to transcriptomic coordinates. | *Rust*

Software Engineer 2017 - 2018
Green Silver Leaves Corp. Tehran, Iran

- Resaa is an online platform for doctor-patient communications, allowing patients to call their doctors using their available credit. I collaborated in developing various features for the administration website and the accounting policy system. | *C#*

Peer-Review Experience

PLOS Computational Biology, PeerJPeerJ Life & Environment
Reviewer

RECOMB 2019 and 2025, ISMB 2019 and 2020, ACM-BCB 2019, APBC 2017
Sub-Reviewer

External Research Support

I contributed to the writing of the grant:

Efficient and scalable pangenomes with the move structures 2/1/2024 - 2/28/2026
NIH/NHGRI R21 grant R21HG013433 ($377K)

Teaching and Mentoring

Mentor for Undergraduate Research Project 2024–2025
Johns Hopkins University, Baltimore, MD

Student: Steven Tan

Instructor of a HEART course Fall 2023
Johns Hopkins University, Baltimore, MD

Needle in a haystack: Finding the origin of shredded sequences

Co-Instructor of a HEART course Fall 2022
Johns Hopkins University, Baltimore, MD

Software Engineering in Bio-medical Research

Teaching assistant of undergraduate courses 2015 - 2017
Stony Brook University, Stony Brook, NY

Social Networks, Discrete Mathematics, Computer Programming III

Teaching assistant of a graduate course Fall 2016
Stony Brook University, Stony Brook, NY

Analysis of Algorithms

Teaching assistant of undergraduate courses 2014 - 2015
University of Tehran, Tehran, Iran

Advanced Programming, Artificial Intelligence

Community Involvement

Conference Scientific Committee May 2025
Johns Hopkins Annual Postdoc Conference

- Provided volunteer support for postdoc conference organization and logistics

Hackathon Judge 2023 and 2024
Hophacks, Johns Hopkins University

- Volunteered as a judge at an annual health-focused hackathon, where participants developed and presented applications over three days in a science fair format.
- Assessed projects on creativity, usefulness, polish, and technical difficulty, contributing to the development and recognition of innovative health technologies.

References

Ben Langmead Associate Professor, Department of Computer Science, Johns Hopkins University
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Rob Patro Associate Professor, Department of Computer Science, University of Maryland
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Travis Gagie Associate Professor, Faculty of Computer Science, Dalhousie University
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Michael I. Love Associate Professor, Department of Biostatistics, University of North Carolina
milove@email.unc.edu